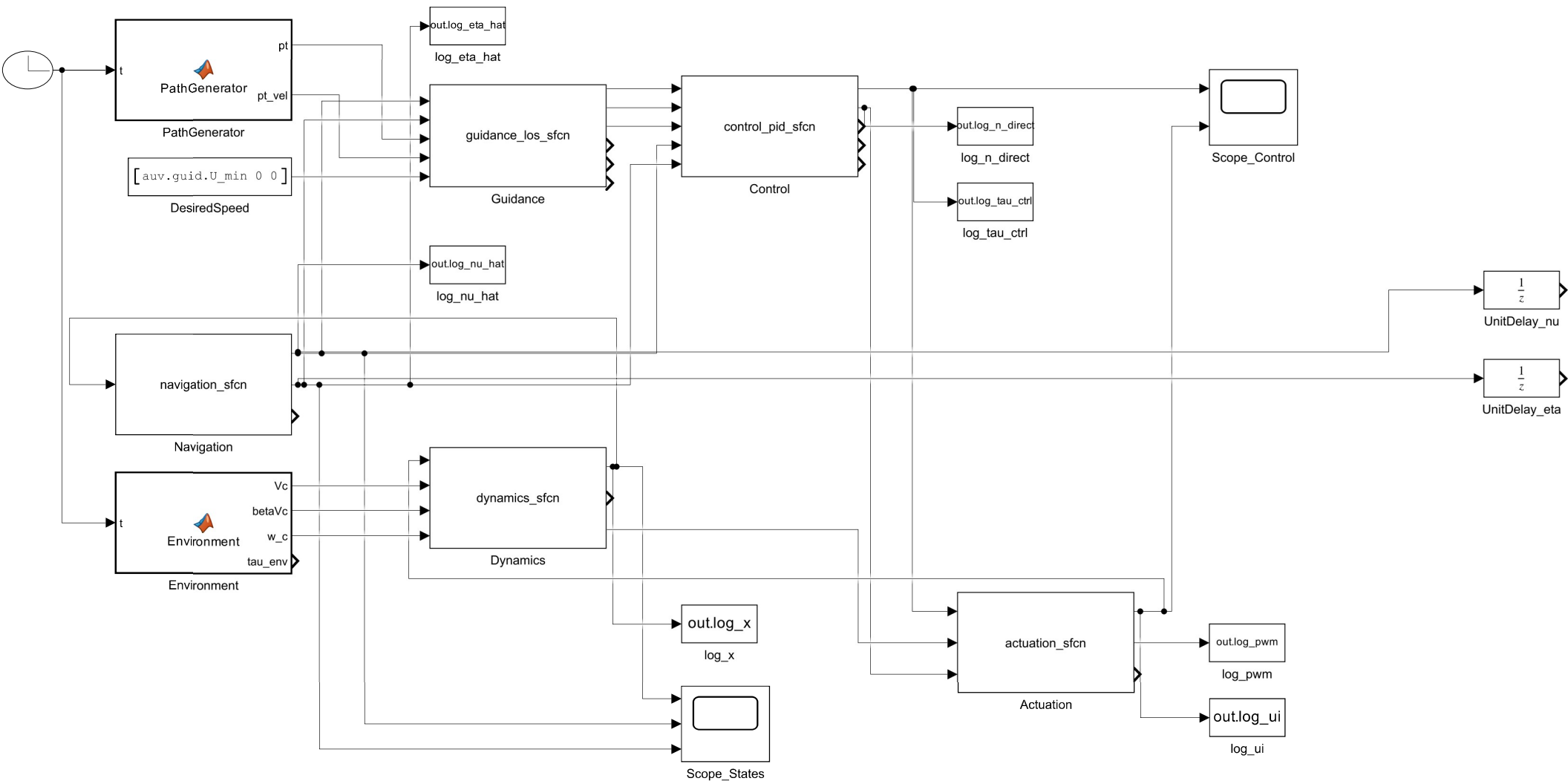


AUV Simulation — Complete 7-Module Architecture
Dynamics: remus100.m (Fossen MSS Toolbox)
Control: PID (swap to SMC: change S-Function name to control_smc_sfcn)
Guidance: 3D LOS
Solver: ode4 fixed-step, Ts=0.010 s



```
function [Vc, betaVc, w_c, tau_env] = Environment(t)
% Reads environment parameters from auv struct in base workspace.
auv = evalin('base','auv');
Vc      = auv.env.Vc_mean;
betaVc  = auv.env.betaVc_mean;
w_c     = auv.env.w_c;
tau_env = zeros(6,1);
```

```

function [pt, pt_vel] = PathGenerator(t)
% Queries the helix path at time t.
persistent path_data
if isempty(path_data)
    auv = evalin('base','auv');
    m=0.6; dt=auv.sim.Ts; T=auv.sim.T_end;
    time=0:dt:T; Yr=m*time;
    X=60*cos(0.02618*Yr); Y=60*sin(0.02618*Yr); Z=2+(2*Yr/200);
    dX=-60*0.02618*m*sin(0.02618*Yr);
    dY= 60*0.02618*m*cos(0.02618*Yr);
    dZ= 2*m/200*ones(size(time));
    path_data.pts=[X;Y;Z]; path_data.vel=[dX;dY;dZ];
    path_data.t_vec=time; path_data.T=T; path_data.dt=dt;
end
t_c = max(0, min(t, path_data.T));
idx = max(1, min(round(t_c/path_data.dt)+1, size(path_data.pts,2)));
pt = path_data.pts(:,idx);
pt_vel = path_data.vel(:,idx);

```

Page	System Name
1	AUV_TopLevel
2 (SF)	AUV_TopLevel/Environment
3 (SF)	AUV_TopLevel/PathGenerator